

Álvaro García Casas

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Education

University Carlos III of Madrid (UC3M)

Bachelor of Science in Biomedical Engineering

May 2027

Georgia Institute of Technology - Exchange

Bachelor of Science in Biomedical Engineering | 4.00 GPA

May 2026

Stanford University School of Medicine

Continuing Medical Education

Sep 2025

Completed 300+ certified hours of advanced online training in medical innovation and research, including AI in healthcare, imaging technologies, and evidence-based clinical methodologies.

Work Experience

R&D Engineer, Georgia Tech Medical Robotics, Atlanta, GA

Aug 2025– May 2026

- Contributing to Limbo, a next-generation prosthetic hand project, a multidisciplinary initiative integration hardware, software and clinical expertise to advance upper-limb prosthetics.
- Characterizing residual muscle activation patterns to optimize electrode placement for proportional myoelectric control in transradial amputees.
- Optimizing signal acquisition and neuromuscular interface strategies while validating electrode configurations to improve control accuracy to over 92% and clinical performance by 15%.

Technical Projects

Co-Founder, Smart Stove Safety Knob (Create-X Venture Accelerator)

Jan 2026 – May 2026

- Led end-to-end development of a retrofit smart stove safety knob in Georgia Tech's Create-X program, turning a \$1,000 seed fund and mentorship into a functional prototype preventing unattended-cooking fires.
- Developed a low-power spring-loaded ratchet mechanism with motion-based shutoff and app-connected monitoring, achieving a sub-\$50 target price and an estimated ~\$15 build cost suitable for large-scale deployment in rental properties.

Wearable Sensor System for Stress and Tremor Analysis in Parkinson's Disease

Aug 2025 – Dec 2026

- Designed and prototyped a wearable forearm cuff integrating surface EMG and GSR sensors to assess how physiological stress affects tremor intensity in Parkinson's disease.
- Developed an Arduino-based EMG/GSR acquisition and MATLAB analysis pipeline for real-time biosignal processing, analyzing 5 trials from 10 patients and detecting stress-related tremor changes ($p = 0.041$) with inverse EMG–GSR correlation ($R^2 = 0.4266$).

Numerical Simulation of Pulsatile Womersley Flow

Feb 2025 – May 2025

- Implemented pulsatile flow simulations in MATLAB, analyzing velocity profiles across a Womersley number (Wo) range of 0.1 to 30 to quantify the dominance of inertial versus viscous forces.
- Validated the numerical model against experimental video data ($Wo = 14.18$) and applied it to an Internal Carotid Artery geometry, calculating a realistic cardiovascular flow Womersley number of 4.15.

System Identification and PID Controller Design for Motor Velocity Control

Feb 2025 – May 2025

- Identified a motor's transfer function in MATLAB to establish a precise model for PID controller design via Root Locus.
- Validated the controller on a physical motor, comparing experimental results against the theoretical model to achieve a target overshoot of 7% and peak time of 0.444 seconds.

Biomechanical Design and Analysis of a Dental Implant using ABAQUS

Aug 2024 – Dec 2024

- Designed multiple dental implant geometries in ABAQUS/CAE, using Finite Element Method to analyze Von Mises stress and minimize axial displacement under a 800 Newton compressive load.
- Optimized the selected geometry to achieve a 4.77 fold mass reduction by hollowing the structure and comparing biocompatible materials to satisfy all yield strength constraints and all displacement under 10 micrometers.

Skills

- Technical:** MATLAB, Simulink, Arduino, Python, LaTeX, CAD, LTSpice, ImageJ, ABAQUS, Microsoft Office, Signal Processing, Analog Electronics, Embedded Programming, EMG, ECG, Mechatronics, PID, Fluid & Solid Mechanics, Electrophysiology.
- Languages:** Spanish (Native), English (C1 Certified)
- Certifications & Awards:** Excellence Scholarship of the Community of Madrid, International Mobility Scholarship, MIT Regenerative AI, Advanced Onshape Assemblies, J&J Robotics, Honorable Mention GT Design Expo, Akademia Future Builders.

Leadership

Mision Pais Spain | Volunteer Team Leader

Jan 2025 - Feb 2025

- Led a team of 30 volunteers on a rural mission trip, supporting individuals with disabilities and promoting community inclusion.
- Organized activities, improved accessibility, and coordinated team efforts to create meaningful social impact.